What is Full Stack Development ?

Last Updated : 20 Nov, 2025

Full Stack Development refers to the development of both front-end (client-side) and back-end (server-side) portions of web applications. They handle the complete development process, from designing user interfaces to managing databases and server-side functions, making them versatile in building end-to-end applications.



Full Stack Development

Technology Related to Full Stack Development

Front-end Development

It is the visible part of website or web application which is responsible for user experience. The user directly interacts with the front end portion of the web application or website.

Front-end Technologies

The front end portion is built by using some languages which are discussed below:

- **HTML:** HTML stands for HyperText Markup Language and is used to design the front-end structure of web pages. Hypertext defines links between pages, while markup language uses tags to define and structure the content.
- **CSS:** CSS, or Cascading Style Sheets, is a simple language used to make web pages visually presentable. It allows you to apply styles to web pages, and importantly, it works independently of the HTML structure.
- **JavaScript:** JavaScript is a famous scripting language used to create the magic on the sites to make the site interactive for the user. It is used to enhancing the functionality of a website to running cool games and web-based software.

Front End Libraries and Frameworks

- **AngularJS:** AngularJS is an open-source JavaScript front-end framework mainly used for building single-page applications (SPAs). It turns static HTML into dynamic HTML, extends HTML with directives, and

HTML, extends HTML with directives, and supports data binding. It is continuously growing and can be freely used and modified by anyone.

- **React.js:** React is a declarative, efficient, and flexible JavaScript library for building user interfaces. It is an open-source, component-based front-end library focused on the view layer and is maintained by Facebook.
- **Bootstrap:** Bootstrap is a free and open-source tool collection for creating responsive websites and web applications. It is the most popular HTML, CSS, and JavaScript framework for developing responsive, mobile-first web sites.
- **jQuery:** jQuery is an open-source JavaScript library that simplifies interactions between HTML/CSS (the DOM) and JavaScript. It streamlines DOM traversal and manipulation, event handling, animations, Ajax requests, and cross-browser development.
- **SASS:** It is the most reliable, mature and robust CSS extension language. It is used to extend the functionality of an existing CSS of a site including everything from variables, inheritance, and nesting with ease.

ease.

- Some other libraries and frameworks are: Semantic-UI, Foundation, Materialize, Backbone.js, Ember.js etc.

Other Important Points

- Work with text editors to use shortcuts and its facilities i.e. Visual studio, Atom, Sublime etc.
- Make UI responsible using grid system.
- Git and git commands like init, add, commit etc for version control and to work with team.
- Other tools like npm & yarn package managers, sass css pre-processor, browser DevTools i.e. chrome devtools.
- Understand using HTTP, JSON, GraphQL APIs to fetch data using axios or other tools.
- It also requires some design skill to make layout and look better.

Back-end Technologies

It refers to server-side development focused on how a website functions. It manages the database through queries and APIs triggered by client-side commands. Such websites consist of the front end, back end, and database, with the back end built using various libraries, frameworks, and

various libraries, frameworks, and languages:

- **PHP:** PHP is a server-side scripting language designed specifically for web development. Since, PHP code executed on server side so it is called server side scripting language.
- **C++** It is a general purpose programming language and widely used now a days for competitive programming. It is also used as backend language.
- **Java:** Java is one of the most popular and widely used programming language and platform. It is highly scalable. Java components are easily available.
- **Python:** Python is a programming language that lets you work quickly and integrate systems more efficiently.
- **Node.js:** Node.js is an open source and cross-platform runtime environment for executing JavaScript code outside of a browser. You need to remember that NodeJS is not a framework and it's not a programming language. Most of the people are confused and understand it's a framework or a programming language. We often use Node.js for building back-end services like APIs like Web App or Mobile App. It's used in production by large companies such as Paypal, Uber

large companies such as Paypal, Uber, Netflix, Walmart and so on.

- **Back End Frameworks:** The list of back end frameworks are: Express, Django, Rails, Laravel, Spring etc.
- The other back end program/scripting languages are: C#, Ruby, REST, GO etc.

Other Important Points

- Structuring the data in efficient way.
- Handle request-response of APIs for storing and retrieve data.
- Security of data is important.

Note: JavaScript is essential for all stacks as it is dominant technology on Web.

Database: Database is the collection of inter-related data which helps in efficient retrieval, insertion and deletion of data from database and organizes the data in the form of tables, views, schemas, reports etc.

- **Oracle:** Oracle database is the collection of data which is treated as a unit. The purpose of this database is to store and retrieve information related to the query. It is a database server and used to manages information.
- **MongoDB:** MongoDB, the most popular NoSQL database, is an open source document-oriented database. The term

11:23 AM 870 B/s
geeksforgeeks.org/blogs
25
DSA Practice Problems C C++ Java

It is a database server and used to manages information.

- **MongoDB:** MongoDB, the most popular NoSQL database, is an open source document-oriented database. The term 'NoSQL' means 'non-relational'. It means that MongoDB isn't based on the table-like relational database structure but provides an altogether different mechanism for storage and retrieval of data.
- **SQL:** Structured Query Language is a standard Database language which is used to create, maintain and retrieve the relational database.

Popular Stacks

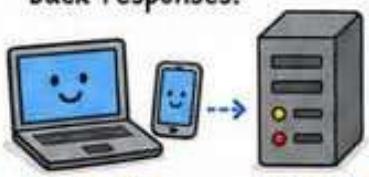
- **MERN Stack:** MongoDB, Express, ReactJS and Node.js
- **MEAN Stack:** MongoDB, Express, AngularJS and Node.js.
- **Django Stack:** Django, python and MySQL as Database.
- **Rails or Ruby on Rails:** Uses Ruby, PHP and MySQL.
- **LAMP Stack:** Linux, Apache, MySQL and PHP.

CLIENT-SERVER ARCHITECTURE

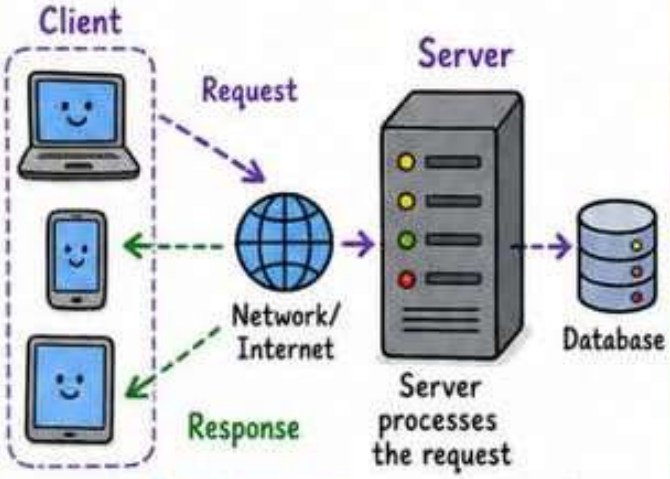
How clients and servers communicate

1. WHAT IS IT?

A client-server architecture is a model where clients request services or data, and servers process those requests and send back responses.



CLIENT ↔ SERVER



alden.social

Clients request • Servers respond

2. EXAMPLE

- Open website
- Browser sends request
- Server finds page/data
- Webpage is returned

Example: Your browser asks for a web page, and the server sends it back.

3. HOW IT WORKS

1. Client sends a request
2. Network carries it
3. Server processes it
4. Server returns a response

4. BENEFITS

- Centralized resources
- Easier maintenance
- Scalable services
- Better security control



5. COMMON EXAMPLES

- Web browsing
- Email apps
- Mobile apps calling APIs
- Online banking & shopping



IN A NUTSHELL

- Client = asks for data or service
- Server = processes and provides it
- ↔ Request goes to server, response comes back
- This model powers most web and app experiences